

Tutorial 3

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)

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1. Find the Supremum and Infimum of the following sets.

(A) $\{\frac{m}{n} + \frac{4n}{m} \mid m, n \in \mathbb{N}\}$

(B) $\{x + \frac{1}{x} \mid x > 0\}$

(C) $\{(-1)^n + \frac{1}{n} \mid n \in \mathbb{N}\}$

(D) $\{\frac{m}{m+n} \mid m, n \in \mathbb{N}\}$

2. Let $a, b \in \mathbb{R}$ and $a < b$ then $a < \frac{a+b}{2} < b$.

3. Prove that for any $a, b \in \mathbb{R}$ this inequality hold $||a| - |b|| \leq |a - b|$.

4. Check whether the following functions are bounded or unbounded.

(A) $f(x) = x^2 - x, \quad D = [0, 1]$.

(B) $f(x) = \frac{1}{x}, \quad D = (0, 1)$.

5. Check whether the functions are convex and concave?

(A) $f(x) = |x|, \quad x \in \mathbb{R}$

(B) $f(x) = -x^2, \quad x \in \mathbb{R}$

6. By using $\epsilon - N$ definition prove that sequence $\{a_n\}$, where $a_n = \frac{n}{n+1}$, converges to 1.

7. By using $\epsilon - N$ definition prove that sequence $\{a_n\}$, where $a_n = 1 + \frac{(-1)^n}{n}$, converges to 1.

8. If $\{x_n\}$ is a convergent sequence in $\mathbb{R}$, what can be said about the convergence of $\{|x_n|\}$ in $\mathbb{R}$?